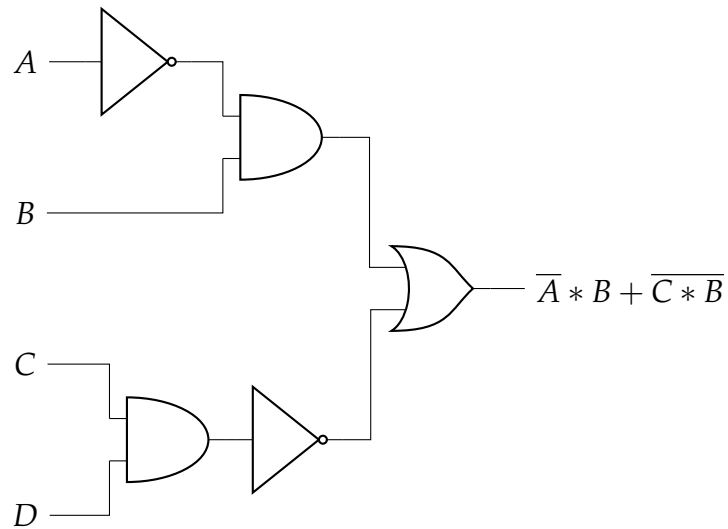


Problem 1

1. $\overline{A} * B + \overline{C} * \overline{B}$



2. Truth Table for MUX

A	B	S	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Problem 2

1. Define Moore's Law in your own words.

It states that the number of transistors on a microchip doubles every two years.

2. Define Dennard Scaling in your own words.

It states that even though transistors get smaller, their power density stays constant.

3. Moore's Law and Dennard Scaling historically enabled rapid advancements in computing by driving exponential growth in transistor density and efficiency. This led to continuous improvements in processing power, energy efficiency, and overall system performance, fueling innovations in everything from personal computers to large-scale data centers.

However, these trends have slowed in recent years. As transistor sizes approach atomic limits, fabrication becomes more complex and expensive, making it harder to sustain Moore's Law. Similarly, the breakdown of Dennard Scaling in the early 2000s led to power and thermal constraints, limiting further increases in clock speed. As a result, modern computing advancements now focus on alternative strategies such as multi-core architectures, parallel processing, and domain-specific accelerators (e.g., GPUs and AI chips).

Problem 3

1. Convert the following numbers to 8-bit binary two's complement representation.

a. $-12 = 1111\ 0100$

b. $101 = 0110\ 0101$

2. Convert the following 12-bit binary 2's complement numbers to decimal.

a. $1010\ 0110\ 1111 = -2048 + 512 + 64 + 32 + 8 + 4 + 2 + 1 = -1425$

b. $0001\ 1101\ 1100 = 256 + 128 + 64 + 16 + 8 + 4 = 476$

3. Convert the following 16-bit binary number to hex.

a. $1111\ 1000\ 0101\ 1010 = \text{F85A}$

b. $0101\ 1010\ 1000\ 0101 = \text{5A85}$

Problem 4

1. B since $\text{nor}(R_0, R_0) = \sim R_0 = \sim 0$
2. B, C, D, E
3.
 - 2's Complement: (D)
 - Fixed Point: (A)
 - IEEE 754: (C)
 - Double Precision: (E)

Problem 5

Initial Register State		Initial Memory State	
Register	Value	Address	Value
X0	0xD	0x00001000	0x44
X1	0x1001	0x00001001	0x12
X2	0x6E	0x00001002	0x44
X3	0x3CB1	0x00001003	0x2E
X4	0x1000	0x00001004	0xA1
X5	0x5	0x00001005	0x72

Initial Register State		Initial Memory State	
Register	Value	Address	Value
X0	0x1002	0x00001000	0x44
X1	0x1001	0x00001001	0x12
X2	0x0	0x00001002	0x44
X3	0x3CB1	0x00001003	0xB1
X4	0xFFFFFFFFB1441244	0x00001004	0x44
X5	0x5	0x00001005	0x12

Problem 6

```
lw 4 5 4    # load from address Gold+4=Gold[4] into r5
add 1 3 7    # add address Go+a=G[a]
add 5 2 6    # add b+Gold[4]
sw  7 6 0    # store at r7+0=G[a] the contents of r6=b+Gold[4]
```

Problem 7

- (a) It will get executed once.
- (b) $r7 = 4$
- (c) $r5 = 3$
- (d) It returns the number of zeros in the array from $[0, Num - 1]$